NFH-Kuza-Google — Unified Domain-Agnostic Architecture

Single platform for Local Services (Agri/Rural) + Health Services + future domains

Domains: **local-services** + **health-services**

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Status: Draft

1. Project Overview

NFH-Kuza-Google is a **unified service orchestration platform** that connects *seekers* (farmers, patients, households) with *providers* (agripreneurs, clinicians, facilities, aggregators) across multiple domains through WhatsApp, a lightweight web application, and low-connectivity channels (SMS/USSD, Mobile Lite). The platform is built on the **Beckn open network protocol**, enabling decentralized discovery and fulfillment of services.

The same underlying flow engine, architecture, technology stack, and data infrastructure serve two live domains today, with a design that admits additional domains without structural change:

- **Local Services / Agri** (Beckn domain code: `local-services`) — Livelihood, Household, Market, and Advisory flows for rural households.
- **Health Services** (Beckn domain code: `health-services`) — example vertical: TB screening, treatment, supervision, and post-TB care (aligned with Kenya's National TB Programme as the pilot).

> **Design Principle**
>
> 100% of seeker interaction occurs through channel-agnostic messaging (primarily WhatsApp) — no app installation required. Providers use a lightweight responsive web application opened via WhatsApp button links, with an optional Mobile Lite interface for field workers (CHVs, CHPs, extension officers) in low-bandwidth conditions.

Service Domains — Agri (local-services)

- **Livelihood Services** — agricultural inputs, equipment rental, veterinary services, soil testing, etc.
- **Household Services** — plumbing, electrical, cleaning, construction, etc.
- **Market Produce** — connecting farmers with aggregators and buyers (reversed economics: buyer pays seeker)

- **Advisory** — agricultural content and guidance delivered as media (no order lifecycle)

Service Domains — Health (health-services)

- **Screening & Referral** — symptom triage, referral to nearest facility with the right capability (e.g., GeneXpert, X-ray)
- **Treatment Adherence** — medication reminders, directly-observed therapy (DOT), supervision by CHP/CHV
- **Lab & Diagnostics** — sputum collection, GeneXpert, culture & DST bookings
- **Post-TB Care & Advisory** — PTLD monitoring, nutrition, TB-awareness education

1.5 Unified Flow Engine

The central abstraction of the platform is a **domain-agnostic 7-step Flow Engine**. Every transactional interaction — whether agri or health — is modelled as the same sequence of states. This is what makes the architecture truly unified: verticals differ in schema and vocabulary, but the engine that orchestrates them is identical.

> **The 7 Steps**
>
> **Need → Search → Select → Confirm → Deliver → Record → Outcome**

- **Need** — a seeker has a need (farmer needs vaccination; patient has persistent cough). User trigger; no Beckn call.
- **Search** — identify who can help, matching capability + location. Beckn: `discover` / `on_discover`.
- **Select** — match to best provider, including quote / booking. Beckn: `select` / `on_select` + `init` / `on_init`.
- **Confirm** — both sides agree, binding contract created. Beckn: `confirm` / `on_confirm`.
- **Deliver** — service rendered (vaccination given; sputum collected). Beckn: `status` updates (`PROVIDER_ACCEPTED`, `IN_PROGRESS`, `COMPLETED`).
- **Record** — data captured (visit notes, clinical record, payment record). Beckn: `update` + `status`.
- **Outcome** — tracked to result (farmer feedback, treatment success, cure confirmed). Beckn: `rate` / `on_rate` plus custom outcome events (e.g., `HEALTH_OUTCOME_RECORDED`). Can be immediate or longitudinal.

Generalized Actor Model

Specific titles (farmer, patient, CHP, agripreneur, clinician) are expressions of six abstract roles. A single individual may play multiple roles across different flows — a CHP is a *Facilitator* when conducting household screening but a *Seeker* when claiming supervision.

Role	Health examples	Local / Agri examples	What they do
Seeker	Household / patient	Household / farmer	Has the need. May be *active* (self-presents) or *passive* (flow initiated on their behalf by a Facilitator or the system).
Facilitator	CHP / CHV	Agent / agripreneur (in facilitation role)	Community-level worker who initiates flows on the seeker's behalf, captures consent, performs first-touch service (screening, needs assessment), and refers to Providers.
Provider	Facility / clinician / nurse / lab	Vet / input supplier / buyer / content provider	Delivers the definitive service, product, or content.
Coordinator	SCTLC / CTLC	Area coordinator / extension coordinator	Supervises Facilitators and Providers across a geography; signs off on supervision flows.
Validator	Lab / mortality auditor	Input-auditor	Provides confirmatory or certifying output required to close a flow.
Sponsor	MoH / Global Fund	Government programme / donor	Funds, commissions, or benefits from aggregate outcomes.

Why Facilitator is a distinct role (not a kind of Provider)

In rural health and agri, the *last mile* is community-based workers (CHPs, agents) who *enable* the service but do not produce the definitive output. A CHP does symptom screening; the facility confirms TB. An agent helps a farmer pick a vet; the vet does the vaccination. Treating Facilitators and Providers as distinct roles gives clarity in provider-matching (`discover`), payment flows (stipends vs fees), accountability (clinical vs first-touch outcomes), and analytics (Facilitator productivity vs Provider throughput).

Domain & Vertical Hierarchy

```
Platform (Kuza Flow Engine)
  └── Domain (e.g., local-services, health-services)
```

```
        └── Vertical (e.g., poultry, crops, TB, HIV, malaria)
              └── Flow Template (e.g., vaccination-booking, screening-referral)
```

Each Beckn domain code maps to one platform Domain. Verticals are sub-classifications within a domain. Flow templates are reusable parameterised patterns instantiated by the Flow Engine.

Capability-Based Matching

Providers are matched by **service area + structured capability tags**, not category alone:

- **Agri**: `services_offered` (vaccination, brooding, AI, soil testing, etc.)
- **Health**: `capabilities` (GeneXpert, X-ray, spirometry, DRTB-certified clinician)

Both stored as structured tags on the Provider profile, queried in `discover`.

Outcome Recording

Beyond service status `COMPLETED`, the engine captures an explicit **Outcome** phase:

- **Agri outcomes**: harvest yield (later season), income earned, repeat business, farmer feedback.
- **Health outcomes**: cure confirmed, treatment success, prevented mortality, PTLD detected.
- Outcomes can be captured *immediately* (feedback at close) or *longitudinally* (months later), feeding back into analytics workloads for ML training.

Flow Initiation Modes

Not every flow starts with a seeker asking a WhatsApp bot. The Flow Engine supports three initiation modes — each with the same 7 steps, only the trigger differs.

Mode	Who triggers	Health example	Agri example
`SEEKER_INITIATED`	Seeker (farmer / patient / household) via chatbot	Patient with cough asks for screening	Farmer requests vaccination for poultry
`PROVIDER_INITIATED`	Provider or Coordinator (CHP / extension officer) — proactive outreach	CHP conducts community screening in a village (Active Case Finding)	Extension officer runs mass vaccination campaign
`SYSTEM_INITIATED`	Flow Engine itself — triggered by outcome	Contact tracing: positive patient's outcome triggers screening flows for	Outbreak quarantine: positive disease diagnosis at one farm triggers vaccination flows at

Mode	Who triggers	Health example	Agri example
	event, scheduled policy, or cascade	household / workplace / school contacts	neighbours within the quarantine radius

> **Same engine, same 7 steps, different trigger**
>
> Regardless of initiation mode, each flow still progresses through *Need → Search → Select → Confirm → Deliver → Record → Outcome*. In provider- and system-initiated flows, the "Need" is registered on behalf of a seeker (who becomes a *passive participant* until they actually meet the provider for delivery).

Cascading & Linked Flows

The **Outcome** step of one flow can become the **Need** of one or more new flows. This is how contact tracing, outbreak response, and scheduled follow-up work.

Every flow records:

- `triggerFlowId` — optional reference to the parent flow that spawned this one (absent for root flows)
- `triggerReason` — why the cascade fired: `CONTACT_OF_POSITIVE`, `OUTBREAK_QUARANTINE`, `SCHEDULED_FOLLOWUP`, `ROUTINE_RECALL`, etc.
- `indexEntityId` (vertical-specific) — the seeker whose outcome initiated the cascade (e.g., index patient UUID in TB contact tracing)

Worked example — TB contact tracing:

```
Flow F1 (SEEKER_INITIATED)
  Patient P walks into a clinic, gets screened, tests positive.
  Outcome: outcomeType = TB_POSITIVE

Flow Engine cascade rule fires:
  For each identified contact of P (household, workplace, school), create a new flow.

Flow F2, F3, F4, ... (SYSTEM_INITIATED, triggerFlowId = F1, triggerReason =
CONTACT_OF_POSITIVE)
  Each contact flow is a screening flow for that contact.
  contactRelationship captured per flow: HOUSEHOLD | COWORKER | CLASSMATE | FRIEND |
NEIGHBOR
  Outbound WhatsApp notification sent to contact (or their household head) informing them of
exposure and scheduled screening.
```

The linkage (F1 → F2, F3, F4) is preserved in the data layer, enabling surveillance, completeness reporting, and longitudinal analysis (e.g., "what percentage of contacts of TB-positive patients actually got screened

within 2 weeks?").

Batch & Campaign Flows

A single provider session can produce many individual flows. One CHP visits a village and screens 40 people — that's **one session, 40 flows**, sharing a common batch identifier.

- `batchId` — groups individual flows into a session (UUID).
- `batchType` — `CAMPAIGN` (mass vaccination / screening), `COMMUNITY_VISIT` (CHP outreach), `CONTACT_TRACING_SESSION` (cluster of cascade-triggered flows executed together), `OUTBREAK_RESPONSE`.
- Each flow in the batch has its own `seeker` and independent `Outcome`; the batch is a reporting / logistics grouping, not a merged flow.

> **Why batches matter operationally**
>
> CHPs / extension officers claim per-session stipends, not per-person. Supervisors review batch completeness (did all 40 scheduled screenings happen?). Analytics measure campaign efficacy (positive-rate per session). The batch abstraction makes all three possible without polluting the individual flow records.

2. Architecture Diagrams

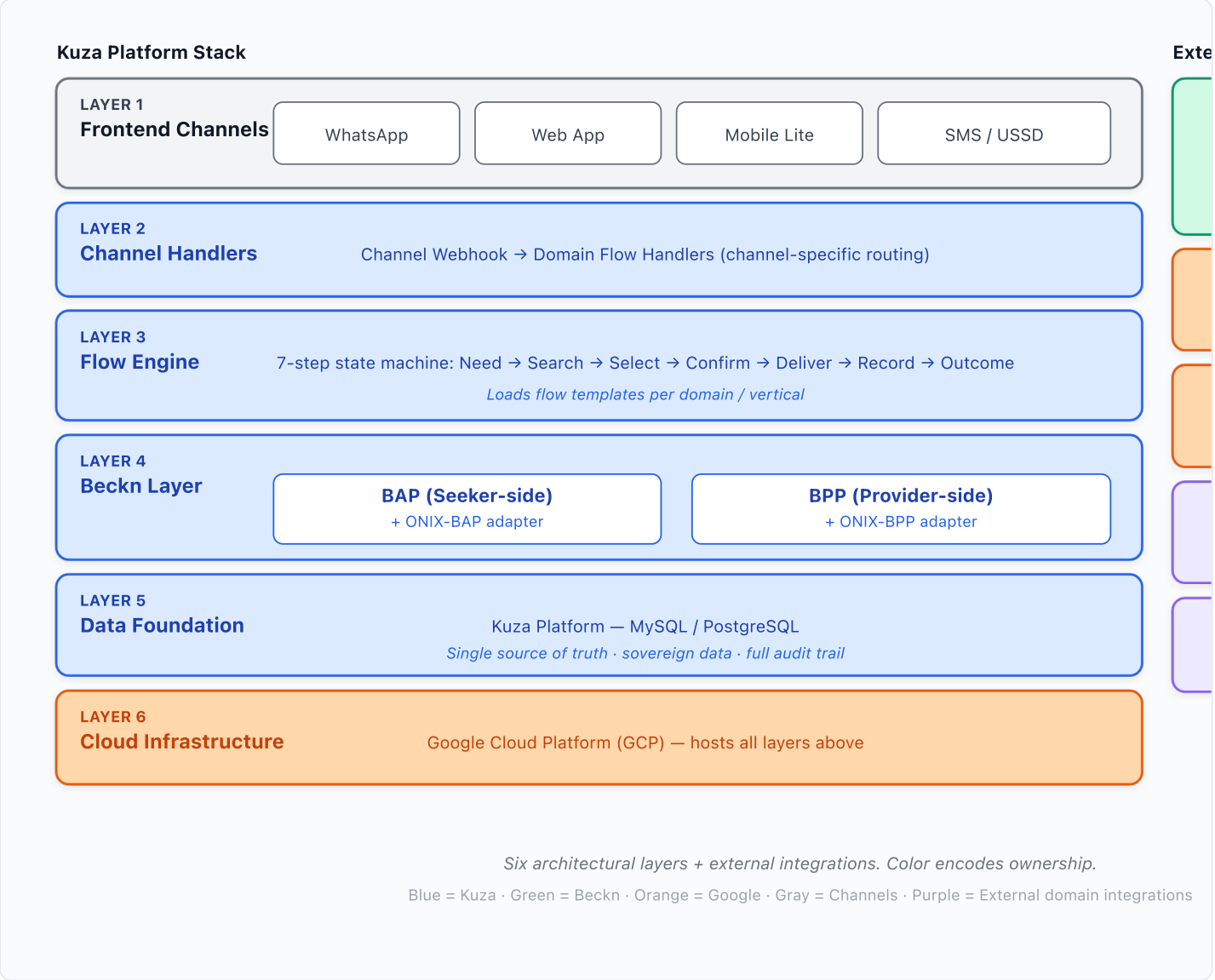
The architecture is presented through three focused diagrams, each answering a different question: what are the layers (A), how does a single request flow through them (B), and what external services integrate with the Kuza stack (C).

Kuza-built components · Beckn infrastructure · Google services · Channels

External integrations (health registers, DHIS2)

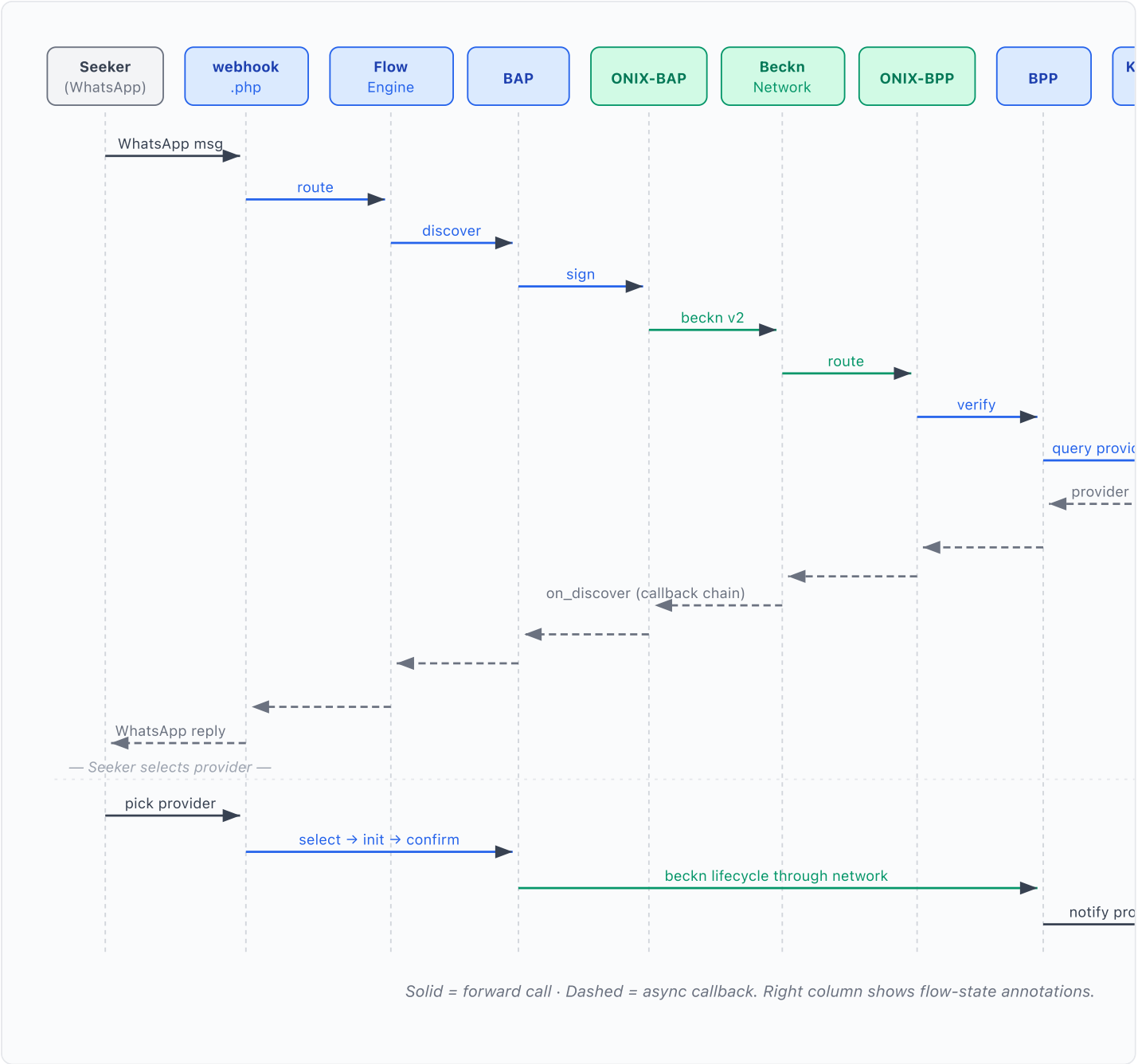
2.1 Diagram A — Layered Stack

Top-to-bottom view of the six architectural layers plus external integrations. No data-flow arrows — just what lives in each layer.



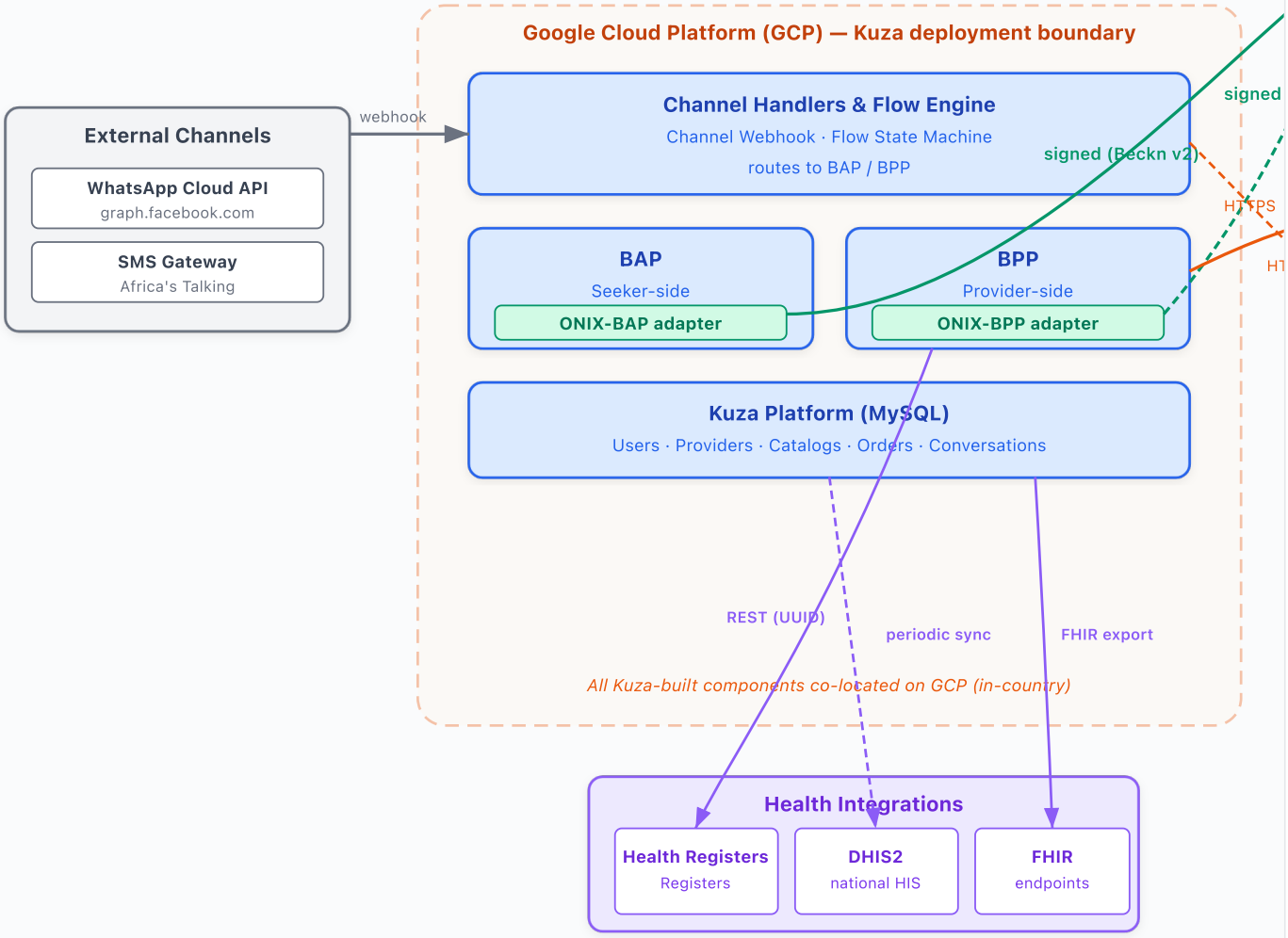
2.2 Diagram B — Request Flow (Sequence)

A single seeker request's journey — discover through confirm — shown with lifelines for each component.



2.3 Diagram C — Network Topology

How the Kuza stack on GCP connects to external services — Beckn Fabric, Google Cloud Services, health integrations, and WhatsApp.



Arrows labelled with protocol. Dashed = async / periodic; solid = synchronous HTTPS.

3. Architecture Layers

The architecture is organised into **six layers** plus a set of external integrations. All six layers run on every deployment; external integrations (Beckn Fabric, Google Cloud Services, disease-programme registers, DHIS2) are consumed as needed per domain.

3.1 Layer 1: Frontend Channels

The platform is **channel-agnostic**. The same flow can be driven over any supported channel; channel choice is a function of the seeker's device and connectivity, not of the domain.

WhatsApp (Seeker-Facing, Primary)

WhatsApp is the primary seeker-facing interface. The bulk of seeker interactions — discovery, provider selection, confirmation, payment declaration, feedback, treatment reminders — occur through WhatsApp messages.

- **Interactive buttons** for small binary/ternary choices (Accept/Reject, Yes/No, payment method selection)
- **Numbered text lists** for longer option sets (provider lists, service categories, sub-categories)
- **Free text input** for location, quantity, symptoms, and other open-ended fields
- Connected via **WhatsApp Cloud API**

SMS / USSD (Low-Connectivity Fallback)

For seekers without smartphones or data, the same flow engine is reachable via SMS and USSD shortcodes. Message structure is simplified (numeric menus, short prompts) but the underlying 7-step sequence is preserved.

Mobile Lite (Field Workers)

A **lightweight mobile client** for field workers (CHVs, CHPs, extension officers) operating in low-bandwidth areas. Supports offline queueing of Deliver / Record steps with sync-on-reconnect.

Kuza Web App (Provider-Facing)

A lightweight responsive web applicationlication opened from WhatsApp button links — no native app installation required. Providers (agripreneurs, clinicians, facilities) use it to:

- Manage their service catalog / capability profile (add/edit services, set pricing, declare capabilities such as GeneXpert, vaccination, etc.)
- View and accept/reject incoming orders / referrals
- Update service status (in-progress, completed)
- Capture clinical / visit records during the Record step
- Manage their profile and service areas

3.2 Layer 2: Channel Handlers

> **Design Constraint**
>
> The existing Channel Webhook Endpoint remains UNTOUCHED. Only routing logic is added to branch marketplace conversations to the new Flow Handler based on trigger words or message IDs.

- **File:** Flow Handler (new PHP file on Kuza Platform)

- **State machine:** manages conversational flow for the 4 marketplace domains (Livelihood, Household, Market, Advisory)
- **Translates** WhatsApp interactions into BAP API calls
- **Receives** async callbacks from BAP and formats WhatsApp response messages
- **Own tables:** conversation state, session tracking — separate from existing poultry chatbot tables (`platform conversation-state tables`, `platform transition tables`)
- **Provider notifications** are triggered via a messaging queue/cron job, NOT through Channel Webhook Endpoint

3.3 Layer 3: Flow Engine

The Flow Engine sits **above BAP/BPP** and **below the Channel Handlers**. It is the domain-agnostic state machine that implements the 7-step lifecycle (Need → Search → Select → Confirm → Deliver → Record → Outcome) and decides which Beckn call (if any) each step translates into.

- **Flow templates** per domain / vertical (`vaccination-booking`, `screening-referral`, `market-sell-produce`, `treatment-adherence`) are loaded at runtime.
- **Transaction state** is persisted per seeker session (`flow_instance_id`) and correlated with the Beckn transaction id.
- **Routing** — each step may produce a BAP call, a BPP-originated event, or a pure UX transition (e.g., the Need step has no network call).
- **Progressive disclosure** — the same template can render differently by actor role (seeker vs. coordinator vs. validator).
- **Optional adoption** — the engine is not required for read-only analytics; existing non-flow interactions (one-off surveys, content pulls) continue to operate independently.

3.4 Layer 4: Beckn Layer (BAP + BPP)

The Beckn layer hosts both the seeker-side participant (BAP) and provider-side participant (BPP), each paired with its ONIX adapter. Both are Kuza-built, hosted on GCP, and communicate with the external Beckn Fabric (see External Integrations, 3.7).

BAP (Beckn Application Platform) — Seeker-side

- Receives translated requests from the Flow Engine
- Constructs Beckn v2 protocol messages: `discover`, `select`, `init`, `confirm`, `status`, `cancel`, `rate`
- Sends protocol messages through the **ONIX-BAP adapter** (Go / Docker)

- Receives async callbacks (`on_discover` , `on_select` , `on_init` , `on_confirm` , `on_status` , `on_cancel` , `on_rate`)
- Pushes results back to the Flow Engine for seeker delivery

BPP (Beckn Provider Platform) — Provider-side

- Receives Beckn protocol requests from the network via **ONIX-BPP adapter** (Go / Docker)
- Connects to the Kuza Platform to fetch provider catalogs, service areas, and pricing
- Manages order lifecycle on the provider side (create, accept, update status, complete)
- Sends async Beckn responses back through ONIX-BPP
- Serves the **Kuza Web App** (provider interface) for catalog and order management

ONIX Adapters

- Plugin-based Go servers running the `fidedocker/onix-adapter` Docker image
- Handle message signing, signature validation, schema validation, and network routing against Beckn Fabric
- Hosted on GCP as Docker containers on the same instance as BAP/BPP application logic

3.5 Layer 5: Data Foundation (Kuza Platform)

Platform relational database (currently MySQL 8, with PostgreSQL migration under way) on GCP serves as the single source of truth. The Beckn network is stateless — all persistent data lives here.

> **Sovereign Data & Compliance**
>
> Seeker / patient / farmer data stays in country. Records are persisted with a full audit trail (status transitions, who updated, when). For the health domain, the record schema is designed to be **FHIR-compatible** so clinical records can be exported into national health information systems without reshaping.

Existing Tables Used

- `user` — unified seeker / provider identity (agripreneurs, farmers, clinicians, patients, CHPs)
- Partner and organization data
- Content library (for Advisory domain, agri & health)

New Tables Required

Table	Purpose
`beckn_provider_catalog`	Services offered by providers, with pricing per service

Table	Purpose
`beckn_provider_service_areas`	County + Sub-County mappings for provider coverage areas
`beckn_subcategory_master`	Dynamic taxonomy of service sub-categories
`beckn_orders`	Order records with full lifecycle tracking
`beckn_order_status_history`	Audit trail of status transitions per order
`beckn_payments`	Payment records (farmer-side and provider-side)
`beckn_feedback`	Feedback and ratings from both parties
`beckn_conversation_state`	Marketplace conversation state machine data
`beckn_farmer_profile`	One-time farmer profile fields (location, preferences)

3.6 Layer 6: Cloud Infrastructure (GCP)

The entire Kuza-built stack is deployed on **Google Cloud Platform (GCP)**. This is the foundational hosting layer on which Layers 1 through 5 run.

- **Compute:** hosts the Channel Handlers, Flow Engine, BAP, BPP, Kuza Web App, ONIX adapters (Docker)
- **Database:** hosts the Kuza Platform data store (currently MySQL 8; PostgreSQL migration planned)
- **Networking:** public endpoints, TLS termination, load balancing for BAP / BPP callbacks
- **Co-location benefit:** Kuza application code and Google Content APIs / AI interfaces are co-located on the same cloud, minimising latency for in-flow API calls
- **Sovereign data:** patient / seeker / farmer data stays in-country (Kenya region) per DPDP-style compliance requirements

3.7 External Integrations

The platform's six architectural layers talk to several external services. These are **not layers** of the architecture; they are third-party systems the platform integrates with.

3.7.1 Beckn Fabric (external Beckn Network services)

- **DeDi Registry** — identity & routing for BAP / BPP participants (`fabric.nfh.global/registry/dedi`)
- **Catalog Service** — BPP publishes catalogs (`fabric.nfh.global/beckn/catalog`)
- **Discovery Service** — handles `discover` requests, queries Catalog Service
- All hosted externally by Beckn Fabric; Kuza connects via ONIX adapters

3.7.2 Google Content APIs — Curated Domain Content

Google publishes **curated content APIs per domain** (agri, health, market, finance, etc.) that serve *informational / educational* content via the chatbot. These are separate from Beckn transactional catalog data — they don't list providers for sale, they deliver knowledge.

- **Primary consumer:** Advisory flow — BPP calls Content APIs during `on_discover` to source advisory material (articles, videos, audio, infographics, structured guides)
- **Secondary consumers:** Livelihood / Household / Market flows — optional enrichment (e.g., "Best practices for vaccinating broilers" attached to a Vaccination order)
- **Called from:** BPP application logic (during request processing); optionally from the Channel Handler for inline content

3.7.3 Google AI Interfaces — Intelligent Processing

Distinct from content APIs, the AI interfaces provide *computational* services that process input and return derived results.

- **Location normalization (all flows):** free-text location → structured County + Sub-County
- **Advisory content tailoring (Advisory flow):** combines Content API output with seeker context to personalise / summarise / translate
- **Future:** intent extraction from free-text queries, language translation, sentiment analysis on feedback
- **Called from:** Channel Handler (location normalization) and BPP (advisory tailoring)

3.7.4 Health Domain Integrations (for `health-services`)

- **Health Registers** — Kenya's National TB Programme data spine (1.3M+ patient records, 23 linked registers). UUID-linked; read / write via REST.
- **DHIS2** — aggregated indicator sync with national / county health systems
- **FHIR R4 endpoints** — optional standard for health data exchange

> **Analytics is a logical capability, not a layer.** Every flow emits event / outcome data into the Kuza Platform (Layer 5). Analytics and ML run as independent workloads consuming this data. See the specification document's *Phased Analytics Roadmap* appendix for how analytics maturity progresses in phases (descriptive → predictive → surveillance).

4. Key Data Flows

Flow A: Farmer Discovers and Books a Service (Livelihood / Household)

```
Farmer (WhatsApp): Selects category --> service --> enters quantity
  --> Channel Webhook --> Flow Handler
    Farmer types location (free text, e.g. "Naivasha")
    --> PHP Handler calls Google AI for location normalization
    <-- Google AI returns {county: "Nakuru", subCounty: "Naivasha"}
    --> WhatsApp: "Nakuru County, Naivasha Sub-County. Correct?" [Yes] [No]
    Farmer confirms [Yes]
    --> BAP Client --> ONIX-BAP --> [Beckn Network / Discovery Service] --> ONIX-BPP -->
BPP
      --> Kuza Platform (query providers by location + category + subCategory)
    <-- BPP --> ONIX-BPP --> [Beckn Network] --> ONIX-BAP --> BAP Client
  <-- Flow Handler --> WhatsApp (list of providers with prices)

(continues through select / init / confirm / status / update / cancel / rate / support —
 see Section 4.6 for the complete Beckn endpoint mapping covering the full order
lifecycle.)
```

Flow B: Provider Receives and Accepts Order

```
Order confirmed in Beckn
  --> BPP --> Kuza Platform (create order record)
  --> Messaging Queue / Cron --> WhatsApp Cloud API --> Provider WhatsApp
    Provider taps [Accept] button in WhatsApp
      --> Channel Webhook --> Provider Routing --> BPP --> Beckn update
    Provider taps [View Details] link
      --> Opens Kuza Web App in browser --> reads from Kuza Platform
```

Flow C: Provider Completes Service, Farmer Confirms

```
Provider (Kuza Web App):
  --> Marks service complete, enters payment details, notes
  --> BPP --> Beckn status update --> ONIX-BAP --> BAP
  --> Flow Handler --> WhatsApp to Farmer
    "Was the service completed?" [Yes] [No]
  Farmer taps [Yes]
  --> Channel Webhook --> Flow Handler
    --> "How did you pay?" [Cash] [M-Pesa]
  Farmer selects payment method
  --> Order marked FARMER_CONFIRMED --> CLOSED
```

Flow D: Farmer Gets Advisory Content (Google Content APIs + AI)

```
Farmer (WhatsApp): Selects Advisory --> Domain (Agri/Health/Market/Finance) --> Category
--> Sub-category
  --> Channel Webhook --> Flow Handler
    --> BAP Client --> ONIX-BAP --> [Beckn Network] --> ONIX-BPP --> BPP

      ┌── BPP processing ──────────────────────────────────────┐
      | 1. BPP calls Google Content API for {domain, category,     |
      |    sub-category} --> returns curated content payload       |
      |    (articles, video URLs, guides)                          |
      | 2. (Optional) BPP calls Google AI Interface to             |
      |    personalise / summarise / translate using farmer        |
      |    context (location, enterprise, language)                |
      | 3. BPP composes the final advisory response                |
      └────────────────────────────────────────────────────────┘

    <-- BPP --> ONIX-BPP --> [Beckn Network] --> ONIX-BAP --> BAP Client
  <-- Flow Handler --> WhatsApp (curated/AI-tailored advisory)

Google Content APIs supply the source material; Google AI tailors it.
Advisory is free, no order lifecycle. Content delivered immediately via on_discover.
```

Domain-agnostic: this flow works for agricultural advisory AND health advisory (e.g., TB awareness, post-TB care education, nutrition counselling). Only the content taxonomy and the Google Content API endpoint change.

Flow E: Market Produce (Farmer Sells)

```
Same flow as A but with reversed economics:
- Farmer has produce to sell (quantity + type)
- discover returns aggregators/buyers with buying prices
- Aggregator collects produce from farmer
- Aggregator pays farmer (not farmer pays provider)
- Both parties confirm and provide feedback

(continues through the same full Beckn lifecycle —
 see Section 4.6 for the complete endpoint mapping including cancellation & feedback.)
```

4.6 Complete Beckn Endpoint Mapping

This section maps every user-facing step in each flow to the corresponding Beckn v2 protocol endpoint, covering the **full order lifecycle** — discovery, selection, initialization, confirmation, fulfilment status, in-flight

updates, cancellation, rating/feedback, and support.

Beckn endpoints reference:

- `discover` / `on_discover` — find services/providers in the network
- `select` / `on_select` — request a priced quote for a specific item
- `init` / `on_init` — submit billing & fulfilment details, receive draft order
- `confirm` / `on_confirm` — create the binding order
- `status` / `on_status` — query or push the current state of the contract
- `update` / `on_update` — mutate an existing contract (e.g. reschedule, report payment, add notes)
- `cancel` / `on_cancel` — cancel an order (before or after provider acceptance)
- `rate` / `on_rate` — submit ratings / feedback on provider or farmer
- `support` / `on_support` — request customer support details

> **Conventions used in the tables below:**
>
> - **BAP → BPP** means the call originates at the BAP (farmer side) and is routed through ONIX / Beckn Network to the BPP (provider side). The corresponding `on_*` callback travels in the opposite direction.
> - **status** calls can be *pull* (BAP → BPP to query) or *push* (BPP → BAP as a notification). Both are used.
> - Status values referenced come from `ServiceFulfillmentAttributes.serviceStatus`: `NEED_REGISTERED`, `SEARCHED`, `SELECTED`, `CONFIRMED`, `DELIVERED`, `RECORDED`, `OUTCOME_RECORDED`, `CANCELLED`, `REJECTED`, `LOST_TO_FOLLOWUP`, `CLOSED`.

4.6.1 Livelihood Services — Complete Lifecycle

Example: Poultry vaccination, feed supply, brooding, incubation, trades & skills.

Phase	User-Facing Step	Beckn Call	Direction	Payload Highlights
Discovery	Farmer picks category, sub-category, service, quantity, location, preferred date	`discover`	BAP → BPP	Intent: category=Livelihood-Services, subCategory, quantityRequired, farmerLocation, scheduledDate
	System returns providers matching service area + service	`on_discover`	BPP → BAP	Catalog with providers, items (ServiceItemAttributes with `verticalAttributes.vertical = "agri-livelihood-poultry"`),

Phase	User-Facing Step	Beckn Call	Direction	Payload Highlights
				offers (ServiceOfferAttributes) filtered by service area
Selection	Farmer selects one provider from the list	`select`	BAP → BPP	providerId, itemId, offerId, quantityRequired
	BPP returns final quote + cancellation policy	`on_select`	BPP → BAP	Final priceAmount (unit price × quantity), cancellationPolicy, confirmed offer terms
Initialization	Farmer provides contact details (auto from WhatsApp profile) and scheduled date	`init`	BAP → BPP	farmer contact, farmerLocation, scheduledDate, billing reference
	BPP returns draft order with payment terms	`on_init`	BPP → BAP	Draft order, payment terms (acceptedPaymentMethods: CASH/MPESA), total
Confirmation	Farmer taps [Confirm] on WhatsApp summary	`confirm`	BAP → BPP	Confirmation with consent flag
	Order created, provider notified	`on_confirm`	BPP → BAP	orderId, provider details, serviceStatus: ORDER_PLACED
Provider Response	Provider taps [Accept] in WhatsApp notification	`status` (push)	BPP → BAP	serviceStatus: PROVIDER_ACCEPTED, confirmedVisitTime
	Farmer receives "Provider accepted. Visit on <date>. Contact: <phone>"	*(push notification, no new call)*	—	Farmer notified on WhatsApp
	Provider taps [Reject] instead	`status` (push)	BPP → BAP	serviceStatus: PROVIDER_REJECTED → CLOSED; farmer re-presented with alternative providers
	Farmer checks order status manually	`status` (pull)	BAP → BPP	Query by orderId; returns `on_status` with current state

Phase	User-Facing Step	Beckn Call	Direction	Payload Highlights
Mid-flight Changes	Provider or farmer changes date, quantity, time slot	`update`	BAP ↔ BPP (either side)	Updated scheduledDate / quantityRequired / confirmedVisitTime
	Other side confirms the change	`on_update`	Opposite direction	Revised order
Service Execution	Provider marks service in progress (via Kuza Web App)	`status` (push)	BPP → BAP	serviceStatus: IN_PROGRESS
	Provider completes service: enters actualServiceDate, payment method, amount received, notes, follow-up flag	`update` + `status`	BPP → BAP	serviceStatus: COMPLETED; providerAmountReceived, providerPaymentReceived: true, providerNotes, followUpRequired; on_update then on_status
Farmer Confirmation	Farmer receives "Was the service completed?" [Yes] [No]	*(WhatsApp; no Beckn call yet)*	—	—
	Farmer taps [Yes] → "How did you pay?" [Cash] [M-Pesa]	`update`	BAP → BPP	serviceConfirmedByFarmer: true, farmerPaymentMethod, farmerPaymentStatus: PAID
	System moves to FARMER_CONFIRMED → CLOSED	`on_update` + `status`	BPP → BAP	serviceStatus: FARMER_CONFIRMED → CLOSED
Cancellation	Farmer cancels before PROVIDER_ACCEPTED (always free)	`cancel`	BAP → BPP	cancellationReason, orderId
	Farmer cancels after PROVIDER_ACCEPTED (within cancellation window)	`cancel`	BAP → BPP	cancellationReason; BPP applies cancellationPolicy (refundable flag, window hours)
	Provider cancels (e.g., cannot service)	`cancel`	BPP → BAP	cancellationReason; farmer re-presented with alternatives

Phase	User-Facing Step	Beckn Call	Direction	Payload Highlights
	Counterparty acknowledges	`on_cancel`	Opposite direction	serviceStatus: CANCELLED → CLOSED; refund details if applicable
Feedback / Rating	Farmer rates provider (1-5) + selects feedback option + optional text	`rate`	BAP → BPP	target: providerId, rating, selectedOption, additionalText (farmerFeedback)
	Provider rates farmer (cooperation, timeliness, payment)	`rate`	BPP → BAP	target: farmerId, rating, selectedOption, additionalText (providerFeedback)
	Acknowledgement	`on_rate`	Opposite direction	Feedback recorded confirmation
Support	Farmer requests provider contact / help	`support`	BAP → BPP	ref: orderId
	BPP returns contact info	`on_support`	BPP → BAP	contactInfo: phone / email / WhatsApp

4.6.2 Household Services — Complete Lifecycle

Example: School fee payment, hospital checkup, insurance enrolment, consumer goods delivery.

Structurally **identical** to Livelihood mapping (4.6.1). The Beckn endpoint sequence and semantics are the same. Category-specific differences are in the payload fields of `on_discover` / `on_select`:

- Same `ServiceItemAttributes` schema — `verticalAttributes.vertical` changes from `"agri-livelihood-poultry"` to `"household-education"` / `"household-health"` etc.
- `subCategory` values: Education, Health, Insurance, Consumer Goods
- `serviceMode` enum: SERVICE_DELIVERY / PRODUCT_DELIVERY / FACILITY_VISIT / DIGITAL
- `beneficiaryType` (INDIVIDUAL / HOUSEHOLD / CHILD / ELDERLY) and `documentsRequired` may be populated
- Some services (e.g., school fee payment) may have `scheduledDate` as the payment deadline rather than a visit date

All lifecycle phases — discovery, selection, init, confirm, status, update, cancel, rate, support — apply the same way. The full Livelihood table above applies to Household with the above payload differences.

4.6.3 Market Produce — Complete Lifecycle (Reversed Economics)

Example: Farmer sells eggs, milk, farm produce to an aggregator / buyer. Money flows provider → farmer.

The Beckn endpoint sequence is the **same as Livelihood**. The semantic reversal is in the payload content, not in the protocol flow. Highlights:

Phase	Beckn Call	Semantic Reversal (vs Livelihood)
Discovery	`discover` / `on_discover`	Farmer advertises produce (`quantityAvailable` , `produceCategory`); BPP returns aggregators/buyers with `buying` prices and `collectionMode` (FARM_GATE_PICKUP / COLLECTION_POINT / FARMER_DELIVERS)
Selection	`select` / `on_select`	Quote is the **buying price per unit** aggregator offers for farmer's produce × quantity
Init / Confirm	`init` , `confirm` (and `on_*`)	Identical protocol, order created to pick up produce at farmer's location / collection point
Provider acceptance	`status` (push)	Aggregator accepts the collection request; serviceStatus: PROVIDER_ACCEPTED
Fulfilment	`status` / `update`	Aggregator collects produce, grades it (`acceptedGrades`), pays farmer; populates `providerPaymentReceived` to indicate *aggregator paid farmer* (NOT farmer paid provider). `amountReceived` = amount paid to farmer.
Farmer confirmation	`update`	Farmer confirms produce was collected and reports method received (Cash / M-Pesa); `farmerPaymentStatus: PAID` = farmer confirms they received payment
Cancellation	`cancel` / `on_cancel`	Either side can cancel before collection. After partial collection, use `update` to adjust quantity actually collected rather than full cancel.
Rate	`rate` / `on_rate`	Farmer rates aggregator's fairness (grading, price honoured, payment promptness); aggregator rates produce quality & farmer reliability
Support	`support` / `on_support`	Same contract as Livelihood

4.6.4 Advisory Content — Discover-Only Lifecycle

Example: Farmer requests advisory on poultry care, financial literacy, health & nutrition.

Advisory is **free, has no order, no payment, no scheduling**. Beckn endpoints used are limited to discovery and optional feedback.

Phase	User-Facing Step	Beckn Call	Direction	Payload Highlights
Discovery with Google Content APIs + AI	Farmer selects Advisory → domain → category → sub-category	`discover`	BAP → BPP	Intent: category=Advisory-Content, subCategory (topic), farmerContext (location, enterprise, language)
	BPP calls **Google Content APIs** (per chosen domain — agri / health / market / finance) to retrieve curated content	*(BPP-internal; not a Beckn call)*	BPP ↔ Google Content API	Curated payload: articles, video URLs, structured guides, infographics
	BPP optionally calls **Google AI Interfaces** to personalise / summarise / translate content using farmer context	*(BPP-internal; not a Beckn call)*	BPP ↔ Google AI	Tailored content matched to farmer language, location, enterprise
	Final advisory delivered to farmer on WhatsApp	`on_discover`	BPP → BAP	Catalog with ServiceItemAttributes (`verticalAttributes.vertical = "advisory-agri"` or `"advisory-health-tb"`): contentType, contentUrl / inline text, contentDuration, targetBeneficiary
Feedback (optional)	Farmer answers "Was this helpful?" [Yes] [No] or rates content	`rate`	BAP → BPP	target: contentId, rating (binary helpful/not or 1-5), optional additionalText
	Acknowledgement	`on_rate`	BPP → BAP	Feedback recorded; may trigger content improvement feedback loop
Support (rare)	Farmer asks for more info / clarification on content	`support` / `on_support`	BAP ↔ BPP	Contact info for advisory provider / content author (if applicable); typically not used in pilot
Not applicable	`select` , `init` , `confirm` , `status` , `update` , `cancel` — advisory has no order, so these endpoints are not used.			

> **Summary of endpoint coverage per flow:**

Flow	discover	select	init	confirm	status	update	cancel	rate	support
Livelihood	✓	✓	✓	✓	✓	✓	✓	✓	✓
Household	✓	✓	✓	✓	✓	✓	✓	✓	✓
Market Produce	✓	✓	✓	✓	✓	✓	✓	✓	✓
Advisory	✓	—	—	—	—	—	—	✓	(rare)

4.7 Domain-Specific Flow Examples

The same 7-step flow engine powers agri and health. Below, the two exemplars side-by-side show that **only vocabulary and payload change** — the sequence, engine, and infrastructure are identical.

Step	Agri — Vaccination	Health — TB Screening
Need	Farmer's flock needs vaccination	Patient has persistent cough > 2 weeks
Search	`discover` near Naivasha	`discover` GeneXpert facility near Kisumu
Select	Pick John Kariuki (vet)	Pick Kisumu County Hospital
Confirm	Confirm 200 birds @ KES 30	Confirm referral, sputum collection date
Deliver	Vaccination performed	Sputum collected, sent for GeneXpert
Record	Provider notes 200 birds, healthy	Lab records test result (FHIR-compatible)
Outcome	Farmer confirms, payment, feedback	TB-positive → initiated on treatment; TB-negative → close

Flow F: TB Screening & Referral (Health domain)

```
Patient (WhatsApp): reports persistent cough via symptom triage
  --> Channel Webhook --> Flow Handler (template: health-services / screening-referral)
    Patient types location (free text, e.g. "Kisumu")
    --> Handler calls Google AI for location normalization
    <-- {county: "Kisumu", subCounty: "Kisumu Central"}
    --> Flow Engine: Search step
    --> BAP --> ONIX-BAP --> [Beckn Network] --> ONIX-BPP --> BPP
      --> Kuza Platform (query facilities by capability=GeneXpert + location)
```

```
    <-- on_discover: ServiceItemAttributes (verticalAttributes.vertical = "health-tb-
screening"; facility, capabilities, hours)
    Patient selects Kisumu County Hospital
  --> select / init / confirm --> order_id generated, sputum collection scheduled
  --> status(PROVIDER_ACCEPTED) --> WhatsApp "Visit on <date>. Bring ID."
  --> Deliver: patient visits, sputum collected
  --> Record: clinician enters test result via Kuza Web App --> status(COMPLETED)
  --> Outcome: HEALTH_OUTCOME_RECORDED
        result=POSITIVE --> auto-triggers treatment-adherence flow (Flow G)
        result=NEGATIVE --> close, advisory content on when to re-screen
```

Flow G: TB Treatment Adherence (Health domain)

```
Trigger: Flow F completes with result=POSITIVE (auto) OR clinician initiates manually
  --> Flow Engine instantiates template: health-services / treatment-adherence
  --> Seeker roles: patient (primary) + CHP (Provider for DOT supervision)

Daily loop (6-month regimen):
  Scheduler cron --> WhatsApp reminder to patient "Did you take today's dose?" [Yes]
[No]
    Patient [Yes] --> update(dose_taken=true) --> Record step
    Patient [No]  --> escalation --> CHP receives DOT visit task
                  --> CHP completes visit in Mobile Lite --> Record step
  Weekly:
    CHP confirms supervision visit, uploads signed DOT form
    Coordinator (SCTLC/CTLC) reviews adherence dashboard
  End of regimen:
    Outcome: CURED / TREATMENT_COMPLETED / LOST_TO_FOLLOWUP / FAILED
    --> HEALTH_OUTCOME_RECORDED event --> analytics consumers

Note: Same state machine, same BAP/BPP, same Kuza data store as any agri flow.
Only the flow template, capability tags, outcome vocabulary, and payload schemas differ.
```

5. Component Ownership

Component	Built By	Hosted On
Channel Handlers (WhatsApp, SMS/USSD, Mobile Lite) — incl. Flow Handler	Kuza	GCP
Flow Engine (7-step state machine + flow templates)	Kuza	GCP

Component	Built By	Hosted On
BAP application logic (agri + health)	Kuza	GCP
BPP application logic (agri + health)	Kuza	GCP
Kuza Web App (provider interface)	Kuza	GCP
Provider WhatsApp notifications	Kuza	GCP
ONIX Adapters (BAP + BPP)	Beckn (open source, Go)	GCP (Docker)
DeDi Registry	Beckn	Beckn Fabric (external)
Catalog Service	Beckn	Beckn Fabric (external)
Discovery Service	Beckn	Beckn Fabric (external)
Google Cloud Platform (GCP) — hosting infrastructure	Google	GCP
Google Content APIs (curated content for agri / health / market / finance)	Google	GCP
Google AI Interfaces (location normalization, content tailoring, future intent extraction)	Google	GCP
MySQL database (with PostgreSQL migration in flight via `kuza-next`)	Kuza	GCP (in-country)
Analytics & Intelligence (logical capability — dashboards, ML models, surveillance)	Kuza	GCP

6. Database Schema

The following schema additions are required on Platform relational database (GCP). These tables live alongside existing Kuza Platform tables and reference the existing `user` table via foreign keys.

Entity Relationship Summary

```
user (existing)
  |-- beckn_farmer_profile      (1:1, farmer-specific fields)
  |-- beckn_provider_catalog    (1:N, provider's services)
  |-- beckn_provider_service_areas (1:N, coverage areas)

beckn_subcategory_master        (standalone taxonomy)

beckn_orders
  |-- beckn_order_status_history (1:N, audit trail)
  |-- beckn_payments             (1:N, farmer + provider payments)
  |-- beckn_feedback             (1:N, ratings from both parties)

beckn_conversation_state        (keyed by phone + session)
```

Key Table Definitions

`beckn_orders`

```
order_id          BIGINT AUTO_INCREMENT PRIMARY KEY
order_uuid        VARCHAR(36) UNIQUE NOT NULL
beckn_order_id    VARCHAR(255)        -- Beckn network order ID
farmer_user_id    INT                 -- FK to user.id
provider_user_id  INT                 -- FK to user.id
category          VARCHAR(100)        -- livelihood | household | market | advisory
sub_category      VARCHAR(255)
service_name      VARCHAR(255)
quantity          DECIMAL(10,2)
unit              VARCHAR(50)
quoted_price      DECIMAL(10,2)
currency          VARCHAR(10) DEFAULT 'KES'
status            ENUM('ORDER_PLACED','PROVIDER_ACCEPTED','PROVIDER_REJECTED',
                       'IN_PROGRESS','COMPLETED','FARMER_CONFIRMED',
                       'CANCELLED','CLOSED')
farmer_location   VARCHAR(255)        -- normalized county/sub-county
provider_location VARCHAR(255)
payment_method    VARCHAR(50)         -- cash | mpesa
created_at        TIMESTAMP DEFAULT CURRENT_TIMESTAMP
updated_at        TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
partner_id        VARCHAR(50)         -- multi-tenancy scope
```

`beckn_conversation_state`

```
id                BIGINT AUTO_INCREMENT PRIMARY KEY
phone_number      VARCHAR(20) NOT NULL
session_id        VARCHAR(100)
current_flow      ENUM('livelihood','household','market','advisory')
```

```
  current_step      VARCHAR(100)
  state_data        JSON                -- accumulated selections, context
  beckn_txn_id      VARCHAR(255)        -- Beckn transaction correlation
  last_activity     TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP
  created_at        TIMESTAMP DEFAULT CURRENT_TIMESTAMP
  partner_id        VARCHAR(50)
  INDEX idx_phone_session (phone_number, session_id)
```

7. Deployment Topology

All Kuza-built components are deployed on Google Cloud Platform (GCP), in-country to preserve **data sovereignty** for patient and farmer records. Beckn Fabric services are externally hosted and accessed over HTTPS.

> **Health-domain deployment note**
>
> For health-services deployments, the Kuza Platform data store exposes a **FHIR-compatible** export adapter so clinical records can be synchronised with national ministry-of-health systems without reshaping. Audit trail (who changed what, when) is retained on every record, and seeker / patient PII never leaves the in-country GCP region.

GCP Instance Layout

Component	Runtime	Port	Notes
Kuza Platform (PHP)	Apache + PHP 7.4	443	Existing platform + WhatsApp Handler + BAP + BPP
Kuza Web App	Nginx (static)	443	responsive web application served as static files
ONIX-BAP	Docker (`fidedocker/onix-adapter`)	5001	BAP-side adapter configuration
ONIX-BPP	Docker (`fidedocker/onix-adapter`)	5002	BPP-side adapter configuration
Google Content APIs	Cloud Run / API Gateway	443	Curated content per domain (agri / health / market / finance) for chatbot delivery
Google AI Interfaces	Cloud Functions / Vertex AI / API	443	Location normalization, content tailoring (summarisation, translation, personalisation)

Component	Runtime	Port	Notes
Platform relational database	MySQL Server	3306	Single database instance

External Dependencies

Service	Endpoint	Protocol
WhatsApp Cloud API	`graph.facebook.com`	HTTPS REST
DeDi Registry	`fabric.nfh.global/registry/dedi`	HTTPS REST
Beckn Catalog Service	`fabric.nfh.global/beckn/catalog`	HTTPS REST
Beckn Discovery Service	`fabric.nfh.global`	HTTPS REST (Beckn v2)

Deployment Note

ONIX adapters run as Docker containers on the same GCP instance. They require outbound HTTPS access to Beckn Fabric endpoints and inbound access from the BAP/BPP application logic on localhost.

8. Domain-Shared vs Domain-Specific

A clear separation between what is *shared across all domains* and what is *bespoke per domain* is what makes adding the next domain (e.g., education, finance, agri-insurance) a matter of configuration rather than re-architecture.

Concern	Domain-Shared (one implementation)	Domain-Specific (per domain)
Flow orchestration	7-step Flow Engine, state machine, transaction correlation	Flow templates (vaccination, screening, treatment-adherence)
Beckn protocol	BAP / BPP / ONIX / Network / Registry / Catalog / Discovery; 4 generic extension schemas (`ServiceItemAttributes`, `ServiceOfferAttributes`, `ServiceFulfillmentAttributes`, `ServiceOrderAttributes`)	Beckn domain code (`local-services` vs `health-services`); `verticalAttributes` content shape per vertical (documented in Beckn Schemas doc)
Channels	WhatsApp, Web App, SMS/USSD, Mobile Lite	— (no variation)

Concern	Domain-Shared (one implementation)	Domain-Specific (per domain)
Data layer	Kuza Platform (MySQL / PG), audit trail, multi-tenancy, partner scoping	Record schemas (FHIR-compatible for health; none required for agri); regulatory / compliance fields
Identity & actors	Seeker / Provider / Coordinator / Validator / Sponsor abstraction; UUID spine	Role vocabulary (farmer / patient; agripreneur / clinician / lab / CHP)
Matching	Service-area + capability-tag matching engine	Capability vocabulary (vet, agrovet vs clinician, GeneXpert, X-ray)
Google services	GCP hosting, AI location normalization, AI tailoring	Content API endpoint per domain (agri / health / market / finance)
Analytics	Event / outcome ingestion pipeline; phased maturity framework	Outcome definition (harvest yield vs cure / mortality); KPIs; dashboards
Master data	Counties, sub-counties, wards (shared across domains)	Sub-category taxonomies; outcome vocabularies

9. Architectural Principles

Seven principles guide the design and evolution of the platform. They were distilled from the health-domain deployment patterns and generalised to apply to all domains:

1. **Preserve, don't replace** — existing data systems (platform databases, disease-programme registers, ministry systems) stay; the platform adds orchestration on top. No rip-and-replace.
2. **UUID is the spine** — every entity (seeker, provider, order, flow instance, outcome) has a universal UUID so records can be linked across systems, domains, and time.
3. **Channel-agnostic intelligence** — the same decision / flow step can be delivered via WhatsApp, Web, SMS/USSD, or Mobile Lite. Channel is a rendering concern, not a logic concern.
4. **Domain-agnostic infrastructure** — the same Flow Engine handles agri and health. Adding a domain is a configuration of templates, schemas, and taxonomies — not new infrastructure.
5. **Sovereign data** — patient / farmer data stays in country, with full audit trail. Schemas are compliant (FHIR-compatible for health; auditable for agri).
6. **Progressive disclosure** — UI complexity matches user role. A seeker sees a short prompt; a coordinator sees a dashboard; a validator sees an audit view. Same underlying data, different surfaces.

7. **Flow Engine is optional until proven** — descriptive analytics can run on existing data without the engine. The engine earns its place as outcome data accumulates and predictive capabilities become viable.